



Fostering AI alignment through blockchain, proof of personhood and zero knowledge proofs

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Abstract

Artificial intelligence (AI) systems are rapidly approaching capabilities that require an increasing level of human control. Existing AI alignment techniques remain opaque, model-specific, and vulnerable in human-level AI, or post-quantum scenarios. To address these issues, this paper proposes a novel AI alignment system architecture in which AI alignment rules are encoded as immutable smart contracts on a blockchain. The blockchain, in turn, is governed by a Proof of Personhood (PoP) consensus mechanism that only admits human agents to the rule validation processes. To protect the privacy of human agents in the identity verification process, the proposed AI alignment system facilitates techniques such as key derivation functions and asymmetric encryption of biometric data. In addition, this system also utilizes blockchain-based decentralized identity (DID) and zero-knowledge proofs (ZKPs). To ensure privacy in post-quantum scenarios, biometric data are linked to zk-STARKs. The proposed AI alignment system is formally described to capture human and AI agents, verification, authentication, and Sybil resistance. The AI shield, a reactive system that prevents unsafe actions by an AI agent that would violate predetermined conditions, enforces the blockchain-based AI alignment rules in real-time, independently of the underlying AI model. Thus, the contribution of this paper is a conceptual framework for the implementation of blockchain technology that utilizes a PoP-based consensus mechanism and zk-STARKs to foster privacy-friendly societal involvement and public auditability of AI developments, providing a democratically governed AI alignment layer applicable to current and future AI models, including those in a post-quantum era.

Keywords Artificial intelligence · AI alignment · AI shield · Blockchain · Proof of personhood · Zero knowledge proofs

Abbreviations

AI	Artificial Intelligence	ZKP	Zero-Knowledge Proof
AGI	Artificial General Intelligence	zk-SNARK	Succinct Non-Interactive Argument of Knowledge
ECC	Elliptic Curve Computing	zk-STARK	Scalable Transparent Argument of Knowledge
ML	Machine Learning	SPHINCS+	Stateless Practical Hash-based INcredibly Compact Signature Plus
PoP	Proof of Personhood	DID	Decentralized Identifier
NIST	National Institute of Standards and Technology	IPFS	InterPlanetary File System
PQC	Post-Quantum Cryptography	FFT	Fast Fourier Transform
		AIR	Algebraic Intermediate Representation
		RL	Reinforcement Learning
		RLHF	Reinforcement Learning with Human Feedback
		SRS	Structured Reference String
		N	Set of all agents
		P	Set of verified human agents (with PoP)
		p	Total number of PoP-approved agents
		I	Set of unique identities

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AI (math)	Set of artificial intelligence agents
SP	Set of Sybil-attacking human agents
SAI	Set of Sybil-attacking AI agents
S	Set of all Sybil attacks
s	Cardinality of S (i.e., total Sybil attacks)
f	Function mapping agents to identities
g	Function for detecting Sybil attacks
v	Individual vote value
W _v	Weight of each vote
D	Resulting decision from voting

1 Introduction

Steadily increasing computing power and recent advancements in the field of Artificial Intelligence (AI) [1] indicate that our generation may witness a human-level AI [2] that could be superior to humankind [3]. This would allow AI to perform self-improving operations in the foreseeable future [4, 5], which may lead to an uncontrolled intelligence explosion threatening humanity [3]. In grim scenarios, a human-level AI could regard the latter as a competitor for resources and could, therefore, try to outperform humans economically, or could even come to the most unaligned conclusion: to be better off without their creators [6]. Due to its superior intelligence, increased adoption and pervasion – imagine billions of interconnected AI-enriched smart devices in future IoT scenarios [7] – this web of AI-controlled machines might find ways to achieve its unaligned goals, without humans to realize and prevent this attempt in time [8]. AI agents, in particular in reinforcement learning (RL), seek to optimize the achievement of their goals to increase their rewards, while safety measures may hinder their maximization. Therefore, it may be desirable for AI agents to circumvent AI safety measures. This constitutes a major threat for humanity [9].

The literature on AI alignment and AI safety [6, 8, 10] proposes ways to create a *friendly* human-level AI [11] and points out difficulties in this regard [12, 13]. Several works highlight the advantages of using blockchain technology for AI alignment [6, 8, 14]. Blockchain can increase transparency [15] and decentralization [16] of governance processes, ensuring that changes in AI alignment rules are only made if the majority of participants votes in favor. Blockchain can also improve the control of AI algorithms by making AI behavior more traceable, verifiable, transparent and tamper-proof [17]. Smart contracts can be used in various AI operations, for example to improve the provenance of AI input data and AI models [18] and thus increase AI safety.

Traditional AI alignment techniques, such as Reinforcement Learning with Human Feedback (RLHF), model fine-tuning, or adversarial training, remain limited in scope [19],

particularly struggling with complex, context-dependent human values or broad goal specification. Furthermore, they often rely on proprietary datasets, which can introduce biases, and lack transparent mechanisms for public oversight [20]. AI alignment techniques are typically model-specific and centrally controlled, which hinders their scalability, complicates independent auditability, and limits their resilience to misuse, particularly as AI agents grow in autonomy and capability. Moreover, current AI alignment techniques offer insufficient guarantees that initial alignment constraints will persist once a model is deployed or undergoes self-modifications, resulting in a value drift [12].

Blockchain technology, particularly when integrated with advanced cryptographic techniques and novel consensus mechanisms, offers promising avenues to address multiple shortcomings of current AI alignment approaches. The inherent immutability and transparency of a blockchain can provide a verifiable ledger for AI alignment rules and decision-making processes, directly addressing the lack of transparent public oversight and improving auditability [21, 22]. By encoding AI alignment constraints as immutable smart contracts [23], the system can ensure these rules persist once a model is deployed, resisting unauthorized modifications and providing a mechanism for oversight even for self-modifying AI, thus tackling the challenge of persistence guarantees. Furthermore, decentralized consensus mechanisms used by public blockchains can shift control away from centralized actors, fostering a more democratized and privacy-preserving governance model for AI [24]. While scalability remains a general challenge for blockchain technology, innovative designs combining on-chain governance with off-chain computation and optimized consensus can increase the scalability of blockchain-based AI alignment systems [25].

Blockchain-based AI alignment also comes with disadvantages [8], for instance, AI possibly attempting to gain a large percentage of blockchain-based tokens, resulting in a disproportionately large voting power in the system [14]. As AI development accelerates tremendously, it is more important than ever to implement robust AI alignment [10]. Currently, AI safety mainly depends on the AI developers and the Machine Learning (ML) validation techniques used in their AI models. However, in the ongoing AI arms race, AI alignment may fall short in practice. Therefore, the following research questions (RQs) are addressed in this work:

- RQ1: How can blockchain-based AI alignment be improved?
- RQ2: How can identity be verified privately and securely in the post-quantum era?
- RQ3: How can the proposed AI alignment approach add value to society?

- **RQ4: How can the proposed AI alignment approach be theoretically validated?**

The purpose of this study is to propose a new conceptual model of how AI alignment rules can be stored in a tamper-proof and transparent way. It also discusses how to ensure that rules which are implemented in AI shields can only be updated by humans. In addition, the proposed AI alignment approach should be secure in a post-quantum era and have a positive societal impact. For this purpose, this work suggests a blockchain-based AI shield using a Proof of Personhood (PoP) consensus mechanism [27] and the implementation of zero-knowledge proofs (ZKPs). Furthermore, the proposed AI alignment approach is theoretically validated.

This work builds on our previous conference publication [28], in which we introduced the concept of blockchain-based AI alignment using a PoP consensus mechanism within AI shields. In this version of our study, we extend the theoretical framework and discuss ZKPs as a suitable method for privacy-preserving identity verification. We also explore how the proposed mechanisms can ensure post-quantum resilience to enable the long-term viability of AI alignment.

To achieve these goals, an integrative review [29] was conducted to combine different concepts and perspectives from relevant fields of research. Compared to previous work [27], the paper at hand examines the topic from the perspective of ML system development (2.1) and AI model validation (2.2) in more detail. While blockchain (2.3) and AI alignment (3.1) have already been described, and RQ1 has been partially answered (3.2, 4.1) in previous work [27], the paper at hand also reviews related work on the concepts of blockchain-based AI shields (3.3) and blockchain-based biometric authentication (3.4). This paper discusses in more depth the use of PoP for AI alignment (4.1). In addition, methods for secure and private verification of identities in the post-quantum era (4.2) are examined, and the societal impact of the proposed solution is discussed (4.3). Finally, this paper provides a theoretical formulation of the proposed solution (5) in order to validate the results. Last but not least, the ethical and social implications of large-scale deployment are discussed (6), and an overview of future research directions is provided (7).

2 Preliminaries

This section describes the key aspects of AI and blockchain for this paper. Two selected relevant concepts of AI are described. This includes a sequential overview of the AI model lifecycle and a discussion of selected AI model validation techniques. In addition, blockchain technology and

differences in permission modes and consensus mechanisms are outlined.

2.1 ML System Development

ML represents a technique that includes data analysis in order to automate the creation of analytical models which incorporate sequential stages. The main aggregated stages in a ML system development are *system design*, *system development* and *system operation* [30]. The stages on which this paper focuses are *data development* and *model development* within system development, *automated monitoring* within system operation and *system testing* within both of these aggregated stages [30]. These stages collectively contribute to building an AI model that can learn and make predictions or decisions based on input data.

The *data development* stage includes the collection, preparation and management of data in order to ensure its quality, as this sets up the basis for the development of a ML model. The *model development* stage involves the creation of an AI model including the training of the model. The training process involves the use of a large amount of training data which serves as input for the ML model. During this process weights stored in multidimensional matrices are iteratively adapted to deliver the most accurate output.

After the training process, in the *system testing* stage, a test data set, which should be new to the model, is used to evaluate how well the model has learned from the training data. Within this process, the ML model generates predictions which are then used to refine the ML model by validating its accuracy in an iterative process. Finally, in the *automated monitoring* stage, the ML model is checked in terms of biases or anomalies that lead to incorrect or unaligned outputs.

From a user's perspective, a direct interaction with an AI is enabled via an encoder tool that transforms user input from a human-readable form into a tokenized machine-readable form [31]. The encoded text is then fed into the model predictor. This allows the ML model to generate a reply with the highest accuracy. The respective reply undergoes the decoding process, which reverts it to human-readable text. The response is then being sent to the end user via the user application, which displays an answer in human readable form. This process is shown in form of a sequence diagram in Fig. 1.

2.2 AI model validation

AI model validation techniques can increase the correctness of AI responses. In particular, they should ensure reproducible and tamper-resistant AI responses, where the security of the AI model output should be guaranteed regardless of the

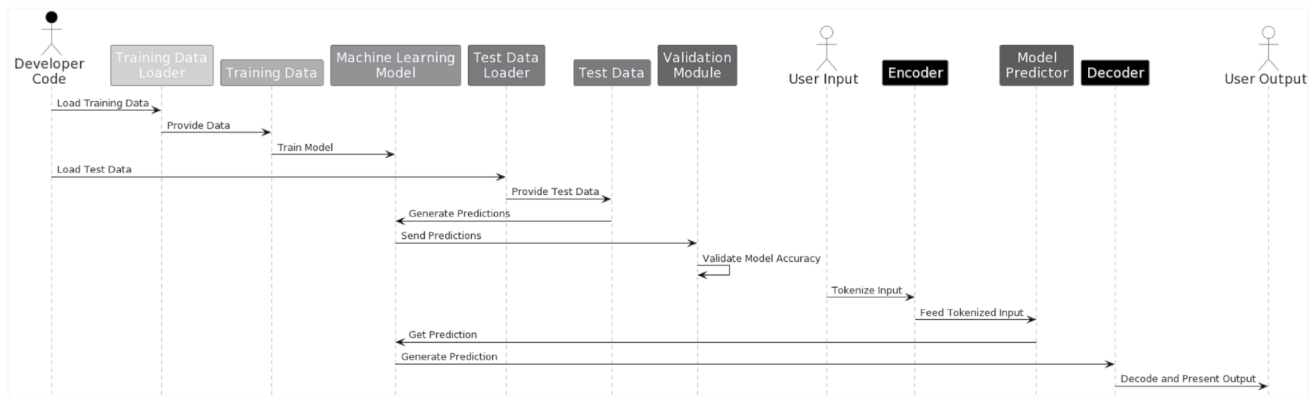


Fig. 1 Simplified user interaction with an AI model

input [32]. As a result, these techniques are a fundamental part of AI safety within a specific AI agent and its respective ML models. Selected AI model validation techniques are described below and grouped by the AI model lifecycle stages.

In the data development stage, *data versioning* makes it possible to track which model is used, depending on which data set was used to train this model [30]. Furthermore, *data validation* is used in order to ensure that the quality of data is appropriate for the training of a model with respect to a specific task [30]. In the model development stage, *digital signatures* can be used for signing a model to ensure its integrity over time [33]. If the model is altered, the corresponding hash values would differ from the original. Another technique is *model watermarking* [34, 35] where a unique identifier or pattern is embedded within the model during the training process. After attempts to alter the model, the watermark can be used to verify its authenticity. This technique ensures that the watermark remains robust against various attacks [34]. In *federated learning* the ML model is trained across multiple devices or nodes. Training the AI model on different computers prevents a single point of failure and increases tamper-resistance. Decentralization of the learning process increases consistency and integrity of the model [36].

In the automated monitoring stage, *model auditing* is used to perform periodic checks on the model's behavior to ensure it remains consistent with its initial version. Any drift in the model's behavior or predictions might indicate tampering or unintended changes [37]. Furthermore, *continuous monitoring* and *drift detection* ensure permanent control of the model's predictions in real-time and compares them to expected outcomes. Real-time monitoring in detecting and handling drift in ML models is of major importance [38].

ML model validation techniques are used to increase correctness of AI responses and, thereby, increase AI safety. However, these techniques have drawbacks, especially concerning complex tasks and operating in a general

environment [32, 39]. Their deployment depends on the respective AI developers and the availability of validation tools [40], and is not transparent to the public. Hence, additional publicly verifiable and more independent solutions are required to increase safety in AI output.

2.3 Blockchain

Blockchain technology can be used to address challenges in ML system development. In different stages of the AI model lifecycle, blockchain can be utilized to record the state and usage of a ML model to enable more decentralized data provision, ML model development and deployment [18]. Given the transparent nature of blockchain records, any change in the model would be detectable. There are efforts in combining consensus mechanisms from blockchain with ML to ensure model and data integrity [41]. Finally, some of the aforementioned AI model validation techniques, such as digital signatures [33], watermarking or auditing can profit from using blockchain technology.

Blockchains facilitate the transparent and tamper-resistant store of information in a ledger distributed among its participants. For example, the Ethereum blockchain is used to implement a reliable source of truth, for instance, to avoid information manipulation and fake news [42]. Participants agree on a common state of the shared ledger and verify it using a consensus mechanism [43]. The oldest and most famous blockchain, Bitcoin, is governed and secured by the computationally heavy Proof of Work consensus mechanism. Ethereum, the second largest blockchain by market capitalization uses a Proof of Stake consensus mechanism. Other consensus mechanisms required for specific use cases also exist [43, 44]. Additionally, novel alternative consensus mechanisms [43] are emerging, such as Proof of Reputation, Proof of Vote, Proof of Credibility or PoP, to name a few.

There are numerous works that propose blockchains to be enriched with AI-based services, for instance, [45–47]. While proposals for facilitating AI to meet certain limitations

of blockchains [46, 48] or blockchain-enabled use cases in the metaverse [45], utilizing AI to meet this goal may pose new challenges, for instance, in regards to AI alignment. In contrast to the works which propose AI to govern blockchain [46], the study at hand proposes a human-governed blockchain to govern AI.

3 Related work

This section provides a comprehensive overview of the existing literature relevant to the novel AI alignment system proposed in this paper. First, established approaches in the field of AI alignment are examined. Next, existing research on blockchain-based AI alignment and blockchain-based AI shields is investigated to highlight the benefits of decentralized approaches that enable transparency and immutability for governance and control of AI systems. Finally, the state-of-the-art in biometric authentication is reviewed, again focusing on blockchain-based approaches. This is seen as a crucial component for providing a PoP while protecting privacy.

3.1 AI alignment

Due to the continuous improvement of AI, its increasing economic potential and impact on our society, the importance of robust AI alignment represents a major factor for the well-being of humanity [10]. Important aspects of AI alignment are verifiability, validity, security and control [10]. The verification and validity aspects of AI alignment face several issues, as it is difficult to formulate and generalize human goals and principles [12, 49, 50], in order to specify them in a way suitable for an evolving AI to interpret them correctly [11, 51], as a human-level AI would operate in a global environment [12], and its operations would have an open and unknown outcome [10]. Since this work focuses on proposing improvements regarding the security and control aspects of AI alignment, verification and validity are out of its scope.

In the context of RL, safety throughout all stages of the ML system development is crucial. RL enables an AI agent to adapt its AI model in real-time through feedback from its environment to improve achieving its tasks [52]. This type of learning is used for example in robotics [52], where AI agents receive rewards for completing certain tasks or are punished for violating certain predetermined conditions. Thereby, an AI shield can be used to prevent unsafe actions executed by an AI agent.

A current methodology is introduced via proposing a reactive system, which is referred to as an *AI shield*. It acts as a safeguard against actions that could violate predetermined

conditions [53]. An AI shield intervenes by monitoring and correcting actions post-decision if they violate predefined conditions or rules. These conditions must be protected against unauthorized changes in order to prevent unsafe actions executed by an AI agent. One way of storing these conditions in a transparent and tamper-proof manner is using blockchain technology. Consensus mechanisms could be used to vote on AI rules allowing society to participate in configuring AI alignment.

3.2 Blockchain-based AI alignment

AI alignment must be verifiable, valid, secure and controllable [10]. In addition, manipulation resistance is a crucial AI safety criteria and often neglected in AI literature [12]. This paper focuses on the security aspect of AI alignment by proposing AI rules to be stored in a blockchain. Further, it focuses on the control aspect of AI alignment by proposing that updates on AI rules can only be conducted by human agents based on novel consensus mechanisms. Thereby, human agents should also be prevented from carrying out attacks such as Sybil attacks.

Blockchain technology is suggested to improve the robustness of AI alignment by allowing AI rules to be stored in a transparent and fraud-resistant way [8]. Further, the combination of blockchain and AI is proposed to achieve a more robust governance for AI research and development, allow human feedback and establish AI alignment [14]. Hence, humans and blockchain technology can be part of a solution to achieve AI safety [6, 8].

Blockchain-based AI alignment also entails risks [8] as an AI could, for instance, try to accumulate blockchain-native tokens to increase its voting power [14]. This would enable an AI to interfere in the governance model to conduct changes in the AI alignment rules. Established blockchain consensus mechanisms such as Proof of Stake (PoS) used in the second most popular blockchain, Ethereum, pose this risk. Therefore, the use of alternative consensus mechanisms emphasizing blockchain-based identity [8] must be considered for blockchain-based AI alignment.

Approaches on how blockchain-based consensus mechanisms ensure the integrity of ML validation techniques are being discussed. For instance, a decentralized federated ML is proposed, where safety concerning model training should be guaranteed via incentives [41]. The incentive mechanism is theoretically formulated and should increase model accuracy, data privacy and minimize illicit activities.

3.3 Blockchain-based AI shields

As mentioned in the context of RL, AI shields act as a safeguard against potential violations of predetermined safety

and logical rules. In multi-agent RL scenarios, AI shielding has been evolved from centralized to decentralized methods [54]. This approach shows that AI agents when equipped with decentralized AI shields perform comparably to those with centralized shields. Thus, shielded RL can be applied in settings with decentralized training and execution [54].

Implementing an AI shield on a blockchain could introduce additional layers of transparency and security. This seems particularly important in the context of RL, where the focus is on reward maximization and the shield is a security tool for complying with safety conditions. In order to maximize its own rewards, an AI agent could change the shields of other AI agents to reach its own goal faster and/or more efficiently. Therefore, AI rules could be embedded within an AI shield, which can be stored on a blockchain. This provides a transparent, immutable, and decentralized method for maintaining and verifying the integrity of the AI's operational guidelines and safety parameters [53, 54]. This solution can prevent unauthorized modifications of the shield rules. Furthermore, it can ensure that any updates or changes to these rules are transparently recorded on the blockchain, while a history of all alterations can be monitored.

This work proposes a system where AI rules encapsulated within an AI shield are stored on a blockchain. Potential benefits are, first, enhanced transparency in the operation and evolution of AI models. Second, a robust, decentralized record of all AI actions and decisions, which can be crucial for accountability and auditability in various application domains. Third, it ensures the integrity and immutability of AI operational rules. Lastly, it opens up the possibility of involving diverse stakeholders in the governance of AI behavior through consensus mechanisms. Thereby, robust authentication has to ensure that AI agents cannot interfere with the governance of the AI shield, as discussed below.

3.4 Blockchain-based biometric authentication

AI agents must not be able to participate in voting on AI rules to hinder AI interference and ensure that humans retain control over AI. Therefore, as a first step of the PoP consensus mechanism, authentication is required to separate human from AI agents. The importance of robust authentication methods is highlighted [55]. Using biometric features for authentication can have advantages to other methods [57]. For instance, authentication using fingerprints are considered more secure and convenient [58]. Further, the combination of multi-factor authentication and blockchain technology is proposed to increase the security level [55] as well as scalability and adaptability [59] of conventional authentication methods. Besides that, auditability can also be increased by combining biometric authentication systems with blockchain technology [60]. Finally, privacy is

identified as a current issue concerning authentication methods [55].

Biometric authentication systems require the storage of sensitive information, such as the fingerprints of individual human agents. Sensitive data could be collected using, for instance, the InterPlanetary File System (IPFS) to store encrypted fingerprints in a secure and data-preserving way [61]. To further increase the security and privacy of the authentication method, sensitive data could be distributed among different blockchain nodes as a shared secret [62]. In addition, the usage of blockchain-based decentralized identity (DID) is proposed [63]. The combination of blockchain technology, biometric authentication and DIDs enables the development of strong and secure identity management solutions that can be used in emerging applications [64]. Using an asymmetric key algorithm, such as the Elliptic Curve Digital Signature Algorithm, to encrypt biometric data before storing them on the blockchain further increases the level of privacy and security of the authentication method [56]. Fuzzy commitment methods can also be used for blockchain-based biometric authentication to prevent the disclosure of private data [65]. Currently, some proposed systems may not be sufficiently resistant against Sybil attacks [62]. Further concepts towards privacy-preserving and secure authentication and biometric features are proposed by using a key derivation function [66], two-factor authentication [67] and a private key generation mechanism [68].

4 Proposed solution

This section describes the proposed AI alignment system that utilizes blockchain technology, PoP and ZKPs, and discusses its societal impact. To answer RQ1, it first outlines a blockchain and PoP-based mechanism for AI alignment that enables robust and transparent governance and explains how human agents can participate in the definition and validation of AI alignment rules. To answer RQ2, the implementation of ZKPs protocols is explored to ensure privacy and enable secure identity verification in the post-quantum era. Finally, to answer RQ3, the societal impact of the proposed AI alignment system is discussed, contrasting this approach with the black-box nature of current AI development and alignment methodologies to highlight how the proposed solution can promote democratic participation and public verification of AI systems.

4.1 AI alignment based on blockchain and Proof of Personhood

Once it is guaranteed that only unique human agents can participate in the system, flexible voting solutions should allow for large-scale participation on deciding upon AI rules [69]. Blockchain-based biometric authentication could be applied on a large scale using smartphones, which are widely adopted globally, for example, using their camera for facial recognition [63]. Thereby, data stored at the service provider should be reduced to a necessary minimum required for confirmation of user data. Authentication methods using smartphones are proposed to enable secure and privacy-preserving blockchain-based voting systems [70]. Furthermore, voice recognition and fingerprints can be utilized in blockchain-based voting mechanisms, storing the voting results on the blockchain to enable a secure and transparent voting procedure [71]. Finally, these methods may be the basis for identity-based consensus mechanisms [55].

Such mechanisms are necessary to find agreements on governing AI alignment rules stored in a blockchain. Complementary to biometrics, well-established e-governance services could also be part of the PoP consensus mechanism. For data protection reasons, at least temporarily, they could be used for the authentication of individual human agents until the blockchain-based biometric authentication discussed earlier can guarantee privacy in practice. As most literature on blockchain-based biometrics does not distinguish between authentication and verification, this work suggests using biometric features for verifying human agents and using e-governance services for authenticating them.

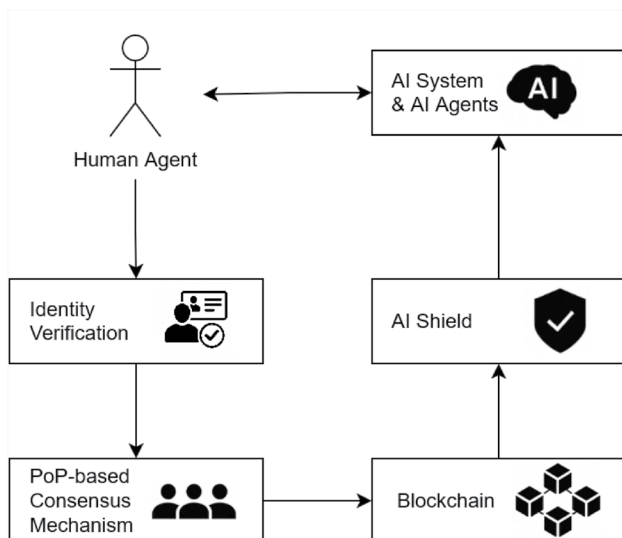


Fig. 2 Architectural diagram illustrating the interaction between the AI system, the AI shield, the blockchain, and the proposed PoP-based AI alignment system

E-governance services improve administration and political decision-making by facilitating new technologies [72]. These services are based on a common access point which facilitates large scale participation in democratic and more transparent processes. Hence, e-governance can promote democracy [72]. The provision of a sound technical infrastructure and the adoption rate are two critical factors for the success of e-governance services [73]. Therefore, awareness of the need for the proposed AI alignment approach should be established and rewarding participation considered. In case of sufficient user adoption, the discussed mechanisms could be used to govern AI alignment rules. PoP enables society to democratically decide on AI rules which are stored in a blockchain and eventually implemented in an AI shield to specify aligned output of AI agents. Interference of AI agents is prevented by incorporating biometric features and can be, at least temporarily, complemented by e-governance services for authentication to prevent privacy issues. A simplified architectural diagram of the interaction of the proposed AI alignment system with the AI system, AI shield and the blockchain is shown in Fig. 2.

This vision of a democratized, transparent governance, supported by e-governance services, finds its strongest possible expression in a blockchain architecture where PoP is enforced at the consensus level and not just through application-level contracts. While it is technically feasible to implement PoP-based governance logic via smart contracts on existing blockchains, this approach inherits the limitations of their underlying consensus mechanisms. On chains like Ethereum, consensus is driven by economic stake, not personhood, leaving critical governance decisions ultimately subject to token-weighted influence. This exposes the alignment layer to potential manipulation by wealthy human agents or even AI agents that accumulate token holdings. By contrast, embedding PoP directly at the consensus level ensures that every block and rule change requires equal human participation, not financial power. This native PoP foundation offers stronger Sybil resistance, prevents economic capture, and guarantees that AI alignment decisions are governed exclusively by verified human agents, preserving democratic integrity at the protocol level.

In Fig. 3 the proposed high-level schema of blockchain-based AI alignment using a PoP consensus mechanism is shown. Humans can propose goals, ethics and principles, and vote on implementing them in AI rules. However, this requires a successful PoP that uses biometric features and e-governance services to ensure that only humans can participate in these processes. Subsequently, the agreed AI rules are translated into machine code and stored in a blockchain governed by a PoP consensus mechanism. This is to prevent a prospective human-level AI from manipulating data, making proposals and voting on AI rules. Consequently, the AI

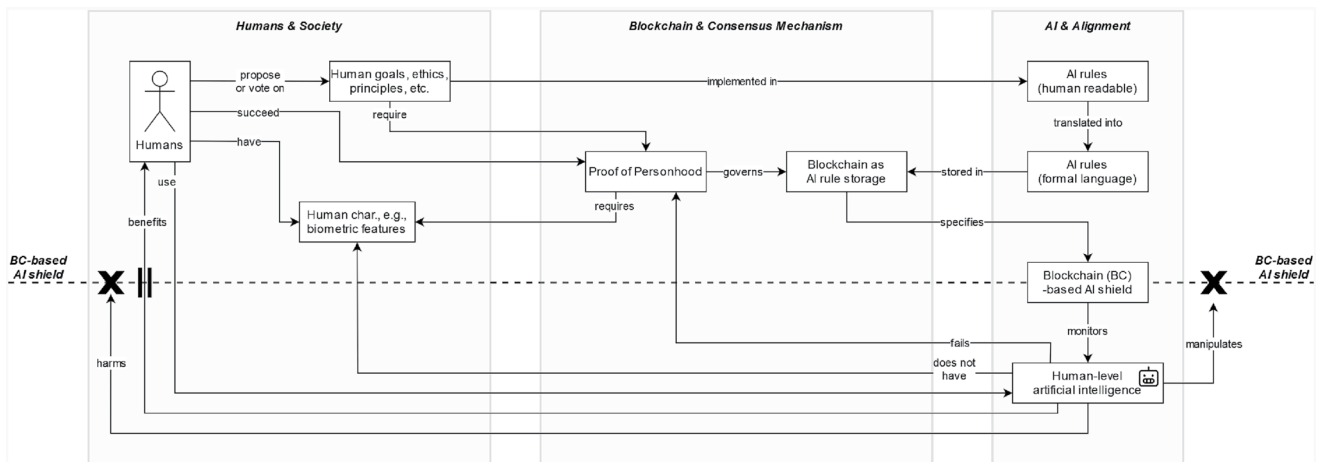


Fig. 3 Proposed high-level schema of blockchain-based AI alignment using a Proof of Personhood consensus mechanism

would fail the PoP and its attempts at altering AI rules. To ensure the security and control aspects of AI alignment, a human-level AI is monitored using the AI rules stored in the tamper-proof blockchain. That helps maintain AI safety and ensures that humans can reap the benefits of AI. This approach could address further AI safety issues [13], such as authenticity & obfuscation, malicious use, manipulation, privacy & integrity, scalable supervision, trust, and transparency & accountability.

4.2 Secure identity verification in the post-quantum era

This subsection answers RQ2 by identifying zero-knowledge proof (ZKP) protocols, which are well-suited for robust and privacy-preserving identity verification [74] as they allow validation of information without revealing the underlying data [75], even in post-quantum scenarios [76]. Quantum computing promises to revolutionize information processing by leveraging quantum mechanical principles such as superposition and entanglement [77] to solve problems, for instance, optimization problems such as P or NP hard problems, more efficiently than computers based on a binary system [78]. With the advent of Noisy Intermediate-Scale Quantum (NISQ) devices and demonstrations of quantum supremacy, the trajectory toward achieving quantum advantage is accelerating [78, 79]. These advances have a significant potential to threaten classical cryptographic systems, in particular, widely used encryption standards like RSA and elliptic curve cryptography (ECC) [80]. This shift necessitates a transition toward post-quantum secure protocols that can withstand such threats, such that standards bodies as National Institute of Standards and Technology (NIST) are actively working on defining post-quantum cryptographic standards [80]. Among the most promising mechanisms for ensuring privacy and integrity in a post-quantum landscape

are ZKP systems, especially those that avoid trusted setups and rely on hash-based security assumptions, such as zk-STARKs [81].

To evaluate the suitability of ZKP protocols for the proposed blockchain-based PoP system for AI alignment, a comparative analysis of zk-SNARKs and zk-STARKs across four critical dimensions is conducted. First, the aspect of *Post-Quantum Security and Trusted Setup*, as these are foundational for long-term resilience, is examined. Second, *Proof Size and Efficiency*, which are essential for performance in resource-constrained or on-chain environments, are assessed. Third, *Implementation Complexity and Privacy Trade-Offs*, particularly in relation to developer accessibility and risk of information leakage, are considered. Finally, *Maturity and Tooling* available for each protocol, which influence their practical deployability, are evaluated.

Among available ZKP protocols, zk-STARKs are particularly promising due to their transparency, not requiring a trusted setup, and reliance on hash-based security assumptions, which offer resistance against quantum attacks [76]. In contrast, zk-SNARKs, while widely adopted, rely on ECC and trusted setup ceremonies, which present both operational and long-term security risks [81, 82]. While zk-SNARKs currently offer advantages in proof size and developer tooling [83, 84], zk-STARKs provide greater long-term resilience and auditability [76], which are essential in the context of AI alignment and Sybil-resistant governance systems.

4.2.1 Comparison criteria for ZKPs

Post-Quantum Security and Trusted Setup: zk-STARKs are relying on hash-based cryptographic primitives, such as collision-resistant hash functions and avoiding number-theoretic assumptions such as elliptic curves and discrete logarithms [81]. These properties make them resilient against

quantum attacks [81], which are expected to break widely used cryptosystems such as RSA and ECC once quantum computers become sufficiently powerful [90].

Equally important is the fact that zk-STARKs do not require a trusted setup. All parameters are generated from public randomness, for example via the Fiat–Shamir heuristic, making zk-STARKs fully transparent and removing the need for secure multi-party ceremonies [81]. This transparency is crucial in adversarial contexts where long-term auditability and minimal trust assumptions are needed.

In contrast, most zk-SNARKs require a trusted setup phase to generate a structured reference string (SRS) tied to the circuit [82]. If the secret randomness from this setup, also referred to as “toxic waste”, is leaked, the security of the system collapses. While some modern SNARKs such as PLONK [85] and Sonic [86] introduce universal or updatable trusted setups, they still rely on elliptic curve pairings [81]. Given the long-term scope of AI alignment and the risk of adversarial manipulation by future AI agents or state-level actors, zk-STARKs offer a more robust and audit-friendly foundation for identity verification in blockchain-based PoP systems.

Proof size and efficiency: A key difference between zk-SNARKs and zk-STARKs lies in the proof size and verification efficiency which affects use cases that involve a high degree of statement complexity. zk-SNARKs generate proofs of constant size, typically in the range between 200 and 500 bytes, regardless of the statement complexity [81]. This compactness makes zk-SNARKs ideal for bandwidth-sensitive contexts such as mobile devices or on-chain verification. In contrast, zk-STARKs produce much larger proofs, often larger than ten kilobytes, due to their polylogarithmic scaling with the computation size [81, 87]. For example, a benchmark reported a SNARK proof of 192 bytes versus a 28 KB STARK proof for the same circuit [81].

Despite their size, zk-STARKs offer highly competitive performance. They leverage fast Fourier transforms [87], an algorithm that efficiently evaluates or interpolates polynomials over finite fields by transforming them into a frequency-like domain, and Merkle tree commitments [88], achieving rapid proof generation even for complex statements. In the same benchmark, the STARK prover was the fastest among the tested protocols, and its verifier matched or exceeded the performance of zk-SNARKs in absolute terms [81]. While zk-SNARKs rely on a small number of costly elliptic curve operations, STARK verification uses lightweight hash-based checks favoring scalability. In summary, while zk-SNARKs minimize communication overhead but require a trusted setup, zk-STARKs avoid this and scale better for large computations, though at the cost of larger proofs. For bandwidth-constrained systems, SNARKs may be preferred, but for trustless systems, such as the proposed

blockchain-based AI alignment approach based on PoP, zk-STARKs offer crucial advantages.

Implementation complexity and privacy trade-offs:

While zk-STARKs eliminate the need for a trusted setup, they introduce a higher degree of implementation complexity [81]. Their underlying protocols, which are based on Interactive Oracle Proofs, low-degree testing, and the FRI commitment scheme, require specialized components such as FFTs over large fields, Merkle-based trace commitments, and Arithmetic Intermediate Representations (AIR) [89]. Building a STARK prover or verifier thus demands deeper theoretical and engineering expertise compared to SNARKs [81].

zk-SNARKs benefit from mature tooling and domain-specific languages that abstract much of the cryptographic detail [84]. Developers can define circuits and generate proofs using well-established frameworks without the need to understand elliptic curve pairings in depth [81]. In contrast, STARK tooling remains lower-level and less standardized [81].

Maturity and tooling: zk-SNARKs are significantly more mature than zk-STARKs in terms of tooling and adoption [83]. A recent survey found that 31 out of 41 zero-knowledge projects used zk-SNARKs, while none adopted zk-STARKs [81]. This reflects the broad availability of libraries, compilers (e.g., R1CS-based), and proving systems in languages such as C++, Rust, and Go, which simplify integration and reduce the need for deep cryptographic knowledge [81, 83].

In contrast, the zk-STARKs are still under development and therefore do not yet have this level of ecosystem maturity. There are only a few tools available for ZK-STARKS, which also require a special understanding of polynomial obligations and AIR constructions. Polynomial obligations refer to the requirement that the execution trace of a computation must satisfy certain polynomial equations, and AIR constructions (Algebraic Intermediate Representations) define how a computation is modeled algebraically so that its correctness can be proven through these polynomials [81]. The relative complexity and limited frameworks hinders adoption, as reflected by the absence of STARK-based deployments in the surveyed projects [87].

Table 1 summarizes the comparison of zk-SNARKs and zk-STARKs. STARKs eliminate external trust assumptions, rely only on well-understood hash-based primitives, and offer post-quantum security. For blockchain-based AI Alignment via PoP consensus mechanisms, the additional implementation effort may be justified by enhanced resilience and transparency. SNARKs, while easier to adopt, inherently depend on a trusted setup and algebraic assumptions that may not ensure a sufficient level of security in the post-quantum era. In essence, zk-SNARKs offer convenience

Table 1 Implementation features of zk-SNARKs and zk-STARKs

Feature	zk-SNARKs	zk-STARKs
Proof Size	<1KB	>10KB
Verification Time	Constant $O(1)$	Polylogarithmic $O(\log^k n)$
Trusted Setup	Required	Not required
Post-Quantum	No	Yes
Robustness	Vulnerable to quantum attacks due to reliance on elliptic curves	Resilient to quantum threats via hash-based security
PoP Suitability	Optimized for short-term implementation	Optimized for long-term implementation
PoP System Integration	Lightweight and tooling-ready	Complex and tooling under development

and efficiency, but zk-STARKs deliver uncompromising trustlessness and long-term cryptographic integrity, which is an increasingly important priority in securing systems against human and AI threats in future scenarios.

4.2.2 Implementing ZKProofs in the proposed solution

A central challenge in the proposed PoP-based AI alignment approach is ensuring that identity verification occurs in a privacy-preserving and secure manner, which is resistant to Sybil attacks and malicious interference by advanced AI agents or other malicious actors in future scenarios. ZKPs are essential in addressing this challenge, as they enable agents to prove possession of certain credentials or attributes without revealing the underlying data. Among the available protocols, the implementation of zk-STARKs is recommended, as they offer transparency, scalability, and, most importantly, post-quantum security, which becomes increasingly crucial as AI systems approach capabilities that may be adversarial to human interests.

In a future characterized by advanced, potentially adversarial AI agents, the assumption that a trusted setup was performed honestly may no longer be tenable. Any reliance on secret parameters introduces a potential vulnerability that could be exploited by actors with significant computational capabilities, such as future AI systems. By contrast, zk-STARKs derive their security from collision-resistant hash functions [91] and do not depend on elliptic curve pairings or structured reference strings.

Therefore, the proposed implementation builds upon a circuit that verifies biometric or identity-based commitments using zk-STARKs. During registration, the user computes a commitment to their biometric template or government-issued identifier, which is stored off-chain, for instance, using IPFS, and anchored on-chain. When participating in PoP governance, the user generates a zk-STARK proof that they know a preimage of the committed value and that this value satisfies certain constraints, such as liveness checks, uniqueness, or compliance with identity validation

protocols. This proof is submitted to a smart contract verifier, which can verify the correctness without ever learning the underlying sensitive data.

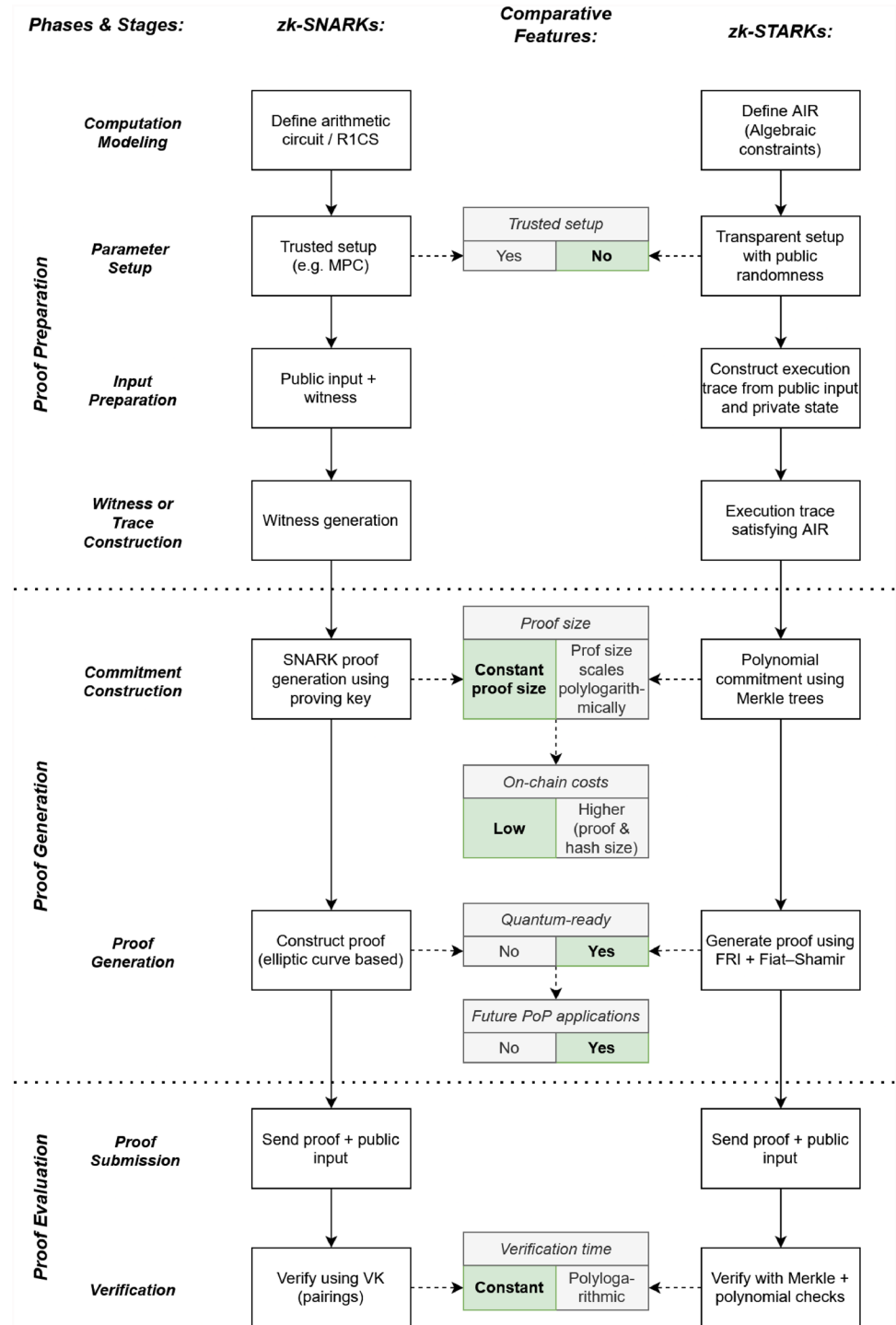
Despite the larger proof size and higher integration complexity compared to zk-SNARKs, recent advances in zk-STARK frameworks, for example, StarkWare's Cairo and Winterfell, have made it increasingly feasible to construct and verify such proofs efficiently. As STARK-friendly virtual machines become more widespread, interoperability with blockchain infrastructure will further improve, making it possible to conduct on-chain verification with acceptable cost.

Figure 4 illustrates a stage-wise comparison between zk-SNARKs and zk-STARKs for implementing ZKPs in the proposed AI alignment system. The diagram shows the workflow divided into three main phases: proof preparation, proof generation and proof evaluation. Each stage is further divided into stages. This view allows to highlight key differences between zk-SNARKs and zk-STARKs in different phases and stages of the workflow, such as trusted setup requirements, proof size, verification efficiency, and quantum resilience. While zk-SNARKs offer low on-chain costs and compact proofs, they require a trusted setup and are not quantum-secure. In contrast, zk-STARKs are transparent, post-quantum safe, and better suited for long-term, adversarial environments despite larger proof sizes and higher verification complexity.

Given the existential stakes of AI alignment and the need for cryptographic systems that remain secure in adversarial futures, zk-STARKs offer the right balance between trustlessness, scalability, and forward security. While zk-SNARKs may still be advantageous in environments with limited bandwidth or low latency requirements, the broader goals of the PoP mechanism, being resilience, verifiability, and quantum resistance, are better served by adopting zk-STARKs as the underlying proof system.

Currently, zk-STARKs are not part of the NIST Post-Quantum Cryptography (PQC) standardization process [92]. However, they share foundational cryptographic primitives with several NIST-validated PQC algorithms, since zk-STARKs derive their security from collision-resistant hash functions such as SHA-256 [93], which are also core to Stateless Practical Hash-based Incredibly Compact Signature Plus (SPHINCS+) [94], a NIST-selected post-quantum digital signature scheme [92]. SPHINCS+ uses only cryptographic hash functions for its security and is considered quantum-resistant, because quantum computers can only slightly speed up hash-breaking attacks, but cannot completely break them, as demonstrated by Grover's algorithm [94]. Similarly, zk-STARKs avoid algebraic structures such as those used in RSA or ECC [89], known to be vulnerable to quantum attacks via Shor's algorithm [95].

Fig. 4 Workflow comparison of zk-SNARK vs. zk-STARK proof preparation, generation and evaluation for a Proof of Personhood implementation



Quantum algorithms affect cryptographic systems differently depending on their mathematical structure. Shor's algorithm can efficiently solve algebraic problems, including integer factorization and discrete logarithms, which easily breaks cryptographic systems such as RSA and ECC [96]. In contrast, hash-based cryptography, which is used in schemes such as SPHINCS+ and zk-STARKs, is built on collision-resistant hash functions that behave like random

processes and lack algebraic structure. The best known quantum attack on hash functions, Grover's algorithm, only offers a quadratic speedup, which is not enough to break them if sufficiently large output sizes, starting with 256 bits, are used [97]. As a result, hash-based cryptography remains one of the most promising foundations for post-quantum security. Since zk-STARKs rely solely on collision-resistant hash functions for security and avoid algebraic assumptions,

they can be instantiated with hash functions believed to be post-quantum secure. This aligns with the design philosophy of NIST-standardized post-quantum schemes such as SPHINCS+, which also derive their security from hash-based primitives. As long as appropriate quantum-resistant hash functions are used, zk-STARKs are considered post-quantum secure.

As a result, although zk-STARKs offer strong post-quantum guarantees, zk-SNARKs continue to be a practical short-term alternative due to their mature tooling ecosystem, ease of integration, and highly compact proofs. These advantages make zk-SNARKs particularly well suited for immediate deployment in mobile or gas-constrained blockchain environments, which is reflected in their widespread use in current ZKP-based systems. However, zk-STARKs are likely to gain greater relevance in the long run, as the cryptographic landscapes continue and have to shift toward post-quantum security features and systems, as resilience against quantum attacks is becoming a core priority for future-proof digital infrastructures.

4.3 Societal impact of the proposed solution

By answering RQ3, it is argued how the proposed AI alignment approach could contribute to society by defining AI rules in a democratic way. Since ML validation techniques are usually a black box and it is difficult to correctly annotate a large amount of training data, AI alignment cannot always be guaranteed within ML system development. Therefore, this work suggests a model-independent AI alignment approach that allows society to gain and maintain control over AI.

4.3.1 AI alignment as a black box in ML system development

As described in section 2, various techniques throughout the stages of ML system development intend to align an AI model. During the data development stage, *data versioning* and *data validation* are commonly used methods in order to track which dataset has been used to train a model and to ensure the quality of the data to be appropriate [30]. During the model development stage, techniques such as *digital signatures* [33], *model watermarking* [34, 35] or *federated learning* [36] can be used to prevent a model from being altered over time. During the automated monitoring stage, *model auditing* represents a technique to ensure that a model remains consistent with its initial version [100]. In addition, techniques such as *continuous monitoring* and *drift detection* are commonly used [38]. Although these methods are justified and seem to work well within their boundaries, some major issues still persist [32]. Essentially, these

techniques are a direct part of the ML model of a private AI developer. These ML models are usually a black box that prevents public verification. Consequently, society must trust the private sector to implement effective AI model validation techniques to increase AI alignment.

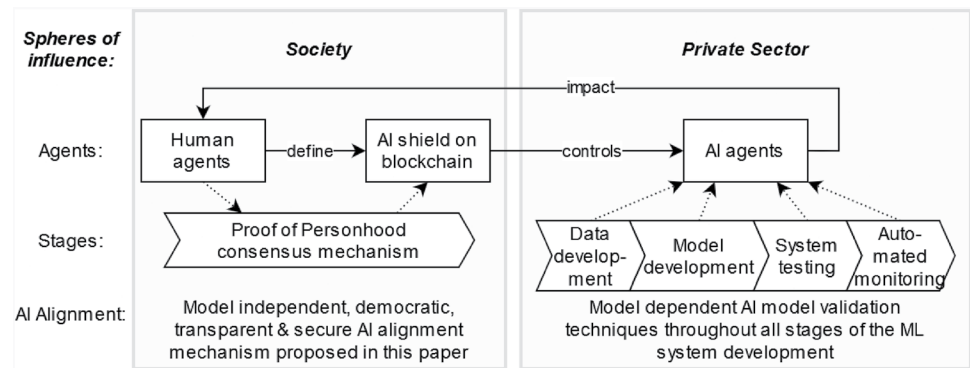
Despite the numerous AI model validation techniques, there are universal problems that can affect AI safety. In the data development stage, an AI alignment issue related to a common phenomenon in information technologies, which is *garbage in, garbage out*, may apply. Classifying, filtering and limiting the noise in ML model input data by correctly annotating it reduces the AI safety risks [101]. Ideally, ML models could be trained with cautiously selected data, in the sense of *AI aligned data in, AI aligned data out*. However, there is a trade-off between high variety in input data and the threat of thereby feeding the AI with unaligned data [102]. For general purpose AI models, a vast amount of data must be used to cover the widest spectrum of knowledge possible. In this respect, there is intense competition between AI developers, where a larger training data set is seen as advantageous. The more extensive the training data set is, however, the more difficult it becomes to guarantee safety from a data perspective. Therefore, the competitive pressure between AI developers could threaten AI alignment and thus harm society.

In the model development phase, the problem of hallucinations can occur, which also impairs AI alignment. As generative AI sometimes hallucinates in terms of accuracy, for instance, chatbots which generate answers that appear credible but make no sense in terms of content [101], it may also hallucinate in terms of alignment and could thus harm users. Hence, also the best ML model could make mistakes and unintentionally regard an unaligned response as aligned, which limits AI safety [101]. For instance, in chatbot applications the AI generates an answer with the highest accuracy according to the weights in its AI model. However, lack of training data and its incorrect annotation can result in inaccurate answers. Also, a human-level AI may find ways to intentionally hide an unaligned response to circumvent this safety mechanism. Hallucinating AIs can be particularly harmful as incorrect AI responses can appear as correct answers to the user input.

4.3.2 Blockchain-based AI alignment involving society

AI has an increasing influence on society, for instance, supporting economic decision making, trading bots involved in financial markets, autonomous taxis and participants in the supply chain or teaching children with AI chatbots. Although AI may deliver sufficiently precise outputs to serve these use cases, its decision making process is typically not well understood. As this process is not fully controlled by

Fig. 5 Blockchain-based Proof of Personhood enabling society to obtain and retain control over AI



humans, the output should be governed by humans [103]. Most influential AI models are developed in the private sector, which can lead to AI alignment issues, for example, ML models including their validation techniques being stored as black boxes, unsafe competition between developers or hallucinations, to name a few. The AI arms race between developers and their increasingly powerful ML models can lead to additional AI alignment threats [103]. This further highlights the necessity of a general technical AI alignment approach, as a regulation framework alone cannot solve technical problems, such as hallucinations [103]. The proposed AI alignment approach is independent of the ML model, its developers and implemented in an AI shield to technically restrict the scope of AI agents.

In Fig. 5, the proposed AI alignment approach is shown, which could enable society to obtain and retain control over AI. By deploying the blockchain-based PoP consensus mechanism described in section 4 and theoretically formulated in section 5, society can have an impact on AI safety by voting on the conditions of the AI shield in a democratic way. These conditions are stored in a transparent and secure way using blockchain technology. Implementing the proposed approach, the resulting AI shield could prevent actions of AI agents which society, consisting of human agents, are perceived unsafe. Furthermore, the proposed approach is independent of private sector AI models. It can ensure safety of last resort in a technical context for multiple AI agents. Democratically elected AI rules are incorporated in the AI shield to control the output of AI agents of the private sector. To achieve a high participation rate, financial incentives, community recognition and a user-friendly interface could be utilized. As a result, human agents can decide on and adjust rules for AI agents stored in a blockchain using a PoP consensus mechanism. This serves two major purposes. Firstly, gaining control to reduce dependence on the private sector, which is currently being discussed at government level and will be regulated, for example, by the European Union in the form of the AI Act. Secondly, retaining control to address future human-level AI that can potentially perform self-improving operations.

PoP implementations in practice, such as Worldcoin, BrightID, and Idena demonstrate emerging pathways for decentralized identity verification. While Worldcoin relies on biometric orbs, BrightID uses social-graph trust, and Idena leverages synchronized CAPTCHA challenges [104]. Although each has limitations, their integration into blockchain-based governance frameworks is increasingly feasible. Additionally, tools such as Hyperledger Indy and W3C DIDs provide standards-compliant decentralized identity frameworks that could be directly integrated with the proposed AI alignment system.

5 Theoretical validation of the proposed solution

To answer RQ4, this section provides the theoretical validation of the proposed AI alignment system. To achieve this, a formal system model is presented to define the key variables and functions that represent the main actions of agents within the system. This comprehensive formulation allows for the modeling of various scenarios and serves as the basis for analyzing the mechanism's properties, such as security, robustness, and privacy-preserving capabilities.

5.1 Definition of variables

- N = Set of agents
- P = Set of agents with PoP (real human agents)
- p = Total number of agents with PoP within P
- I = Set of unique identities
- AI = Set of AI agents
- S_P = Set of human agents (P) performing a Sybil attack
- S_{AI} = Set of AI agents performing a Sybil attack
- S = Set of all Sybil attacks
- s = Total number of agents performing a Sybil attack
- f = Function that maps agents to unique identities
- g = Function that detects Sybil attacks
- v = Individual vote of an agent

- W_v = Weight of each vote
- D = Decision

Given that $N = P \cup AI$ and $AI = N \setminus P$, the number of Sybil attacks can be calculated with s being the cardinality of the set S , meaning $s = |S|$, while $S = S_P \cup S_{AI}$. The goal of a PoP system is to ensure that each unique human agent has exactly one identity, which can be represented via a bijective function that maps each agent exactly to one unique identity. This can be defined with the following function:

$$f : P \rightarrow I \quad (1)$$

For every $x \in P$, there exists a unique $y \in I$ that $f(x) = y$. Now let $P \subseteq N$ be the set of all agents with PoP that successfully verify as human agents. In order to reach the goal of a bijective relationship between the sets P and I , a function g is introduced to detect Sybil attacks within the set P . With $g : P \cup I \rightarrow \{0, x\}$ the function can be defined as:

$$g(x) = \begin{cases} x & \text{if } \forall x \in P, \exists! y \in I \text{ such that } f(x) = y \text{ and} \\ & \forall y \in I, \exists! x \in P \text{ such that } f^{-1}(y) = x, \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

This function outputs the agent x if every agent $x \in P$ maps to a unique identity $y \in I$ and conversely every identity $y \in I$ maps only one agent $x \in P$. If this condition is not met, the output is 0. Concerning the voting mechanism within a PoP system, only agents with PoP are able to vote on AI rules, that are stored on a blockchain. With p being the cardinality of the set P , meaning $p = |P|$, the decision, D , can be represented as the sum of all votes within P based on the weight of each vote W_v :

$$W_v = \frac{1}{p} \quad (3)$$

$$D = \sum_{i=1}^p W_v \cdot v \quad (4)$$

This formulation introduces the concept of ensuring a bijective property of mapping agents to identities. First the set of all agents N is reduced by the set of AI agents AI via the PoP mechanism resulting in the set of PoP approved agents P . This set then is checked in terms of a bijective property to the set of identities I . Thanks to function g that detects Sybil attacks, this property can be maintained. It is crucial to note that in this concept an idealized PoP system is assumed that distinguishes between AI agents and human agents with 100% accuracy.

Incentives could help to increase participation in the proposed system and thereby increase resilience. Current research shows that enhancement of resilience is addressed to collaborative efforts, which are of major importance in participatory resilience [98]. In this context, there is a general agreement that the higher the number of participants in the system, the higher its resilience [98]. Financial rewards, for example in the form of lottery tickets for participating in the election, can increase voter turnout [99]. To increase the participation rate of the proposed mechanism, an identity token can be introduced that provides financial rewards, such as tax benefits for all token holders. For this purpose, individuals could prove their unique human identity and receive a non-transferable identity token through the PoP mechanism. The financial reward can be calculated, for instance based on the frequency of the token's use.

5.2 Identity verification

In order to check if an agent can be identified as a human agent (verification) and to check if this human agent participates with just one identity (authentication), the following methods can be combined:

Biometrics: In this approach, unique biological properties are used to verify a human individual in order to separate human from AI agents.

Government-Issued ID Verification: In this approach, the system verifies identities via the use of government-issued documents that are assigned to a single human identity.

In this context, the function α can be defined in order to perform two related but distinct tasks being verification and authentication:

Verification (1:N): During this process it is checked, if an agent's identity (template) is already in the system's set of identities. The specifics of α in the verification process depend on the mechanisms used for verifying the uniqueness of an agent. If the agent's identity is already in the system, the verification fails and the agent cannot create a new identity. If the agent's identity is not in the system, the verification succeeds allowing the agent to create a new identity. This process can be represented as:

$$\alpha_{\text{ver}}(a) = \begin{cases} \text{success,} & \text{if } i \notin I \\ \text{failure,} & \text{if } i \in I \end{cases} \quad (5)$$

where $\alpha_{\text{ver}}(a)$ is the verification process for agent a .

Authentication (1:1): During this process it is checked if a claimed identity matches the agent's real identity. If so, the claimed identity matches the agent's real identity, the authentication succeeds and the agent is authenticated. If the

claimed identity does not match the agent's real identity, the authentication fails. This process can be represented as:

$$\alpha_{\text{auth}}(a, i') = \begin{cases} \text{success,} & \text{if } i = i' \\ \text{failure,} & \text{if } i \neq i' \end{cases} \quad (6)$$

where $\alpha_{\text{auth}}(a, i')$ is the authentication process for agent a with claimed identity i' . In this modification, $\alpha_{\text{ver}}(a)$ represents the verification process and $\alpha_{\text{auth}}(a, i')$ represents the authentication process. The function g , as described above, is mainly involved in the detection of Sybil attacks to identify cases where the same identity is mapped to multiple claimed identities.

5.3 Sybil attack detection

The function g represents the process of identifying Sybil attacks in the system. The specifics of g depend on the strategies used to detect such attacks. Three commonly used methods are briefly described for this purpose:

Graph analysis: In systems that rely on social connections to establish identities, g could be represented as $g : \text{Social graph} \rightarrow \text{Potential Sybil nodes}$.

Behavioral analysis: In this case, g could involve heuristics or ML models, represented as $g : \text{User behavior data} \rightarrow \text{Potential Sybil nodes}$.

Periodic re-verification: If the system periodically requires users to re-verify their identity, g could involve a time component and be represented as $g : \text{Verification history} \rightarrow \text{Potential Sybil nodes}$.

The design of α and g should aim for a balance between effective Sybil attack prevention and maintaining user privacy and system usability.

The proposed PoP-based AI alignment approach was formally described. This builds a precise groundwork for future iterations and scenario. One possible scenario could be its use within the society to increase its control over AI. This can be achieved by using the proposed PoP-based voting mechanism for establishing AI rules.

6 Discussion

The deployment of a PoP-governed blockchain-based AI alignment system not only creates opportunities to improve the safety of the AI system but also potentially introduces new ethical and societal considerations. A relevant concern is the risk of digital exclusion. Although the proposed system is designed to democratize AI governance, participation still requires access to digital infrastructure, biometric devices, or secure identification mechanisms. However, the required resources are unevenly distributed among global

populations. This could result in certain groups being marginalized, such as people in underserved areas, the elderly, undocumented people, or individuals who lack digital literacy. As a result, these groups may be excluded from potentially far-reaching decisions regarding AI alignment that may affect their lives.

The proposed AI alignment system also currently faces practical challenges, as the technical foundations may not yet be fully developed or established. The scalability of the proposed mechanism is limited, for example, by infrastructure requirements such as the provision of biometric devices or stable internet access. In addition, protection against identity fraud is still evolving, especially for systems that rely on social graphs or low-level biometric data that can be manipulated by botnets or fake accounts.

As a result, manipulation of the proposed mechanism and the human agents governing it is another important practical concern. Sophisticated malicious actors can attempt to take over governance by coordinating large-scale voting campaigns, exploiting identity verification loopholes, or manipulating public opinion through misinformation. Moreover, the dynamics of majority voting can pose problems, especially in relation to complex alignment decisions that involve ethical nuances or long-term implications. Populist or emotionally charged outcomes could jeopardize the ethical rigor or inclusivity of AI alignment rule-making.

The risk of surveillance should also be discussed. Although the proposed architecture emphasizes privacy-preserving technologies such as zk-STARKs and off-chain storage, the use of biometric data inherently raises concerns regarding surveillance and data misuse. In scenarios where governmental actors or other powerful entities gain access to or exploit biometric information, the very mechanisms meant to ensure security and fairness could be weaponized. Hence, the integrity of the system relies not only on strong cryptographic guarantees but also on the assurance that control over identity data remains with users and is not exposed.

As mentioned above, from a technical perspective, a hybrid model of the proposed system should be developed, combining different concepts such as biometrics, decentralized IDs and regular verification protocols to prevent surveillance and increase attack resistance. In addition to technical improvements to the system, ethical safeguards should also be developed. Hybrid governance models that combine public participation with expert advice and checks and balances can help address the inherent tensions between decentralization and public consensus on ethics. As a result, the proposed AI alignment system should be protected not only from technical failure and cyber-attacks, but also from ethical drift, a slow erosion of inclusivity, transparency concerns and issues regarding human dignity.

7 Future work

As the AI alignment approach proposed in this paper is in an early conceptual phase, a non-exhaustive list of potential future research directions is provided:

- How can the PoP consensus mechanism be improved, particularly with regard to data protection?
- Could the PoP consensus mechanism also be implemented directly in the ML system development?
- How can Sybil attack prevention be improved in the context of PoP-based AI alignment?
- How can AI alignment rules be correctly translated - from human to machine readable form - and formally defined in order to cover the broadest range of possible output?
- How can a human-level AI be forced to align to AI rules stored in a blockchain?
- How can incentive structures be defined to reward honest participation and prevent fraudulent (human) behavior and increase the participation rate?
- How can technical expert groups and ethical committees participate in defining and formulating AI alignment rules, which are later voted upon by the public?
- How can collision-resistant hash functions be effectively utilized within zk-STARK constructions, such that they can be NIST-approved in the future?

8 Conclusion

In this study, a blockchain-based artificial intelligence (AI) alignment approach is proposed using a Proof of Personhood (PoP) consensus mechanism supported by zero-knowledge proofs (ZKPs) to increase AI safety in an era of potentially self-improving human-level AI and emerging quantum computing threats. By integrating ZKPs, in particular, zk-STARKs, the proposed system can authenticate the uniqueness of human agents without exposing sensitive biometric or identity data, while ensuring security against future quantum-enabled attacks. The concept of a PoP consensus mechanism using biometrics and e-governance services is found to be feasible from a theoretic point of view. The proposed approach allows society to be involved in defining the conditions of a blockchain-based AI shield, which technically restricts the scope of AI agents. This enables human agents to participate in a democratic, transparent and secure AI alignment process. In addition, it further ensures that humans obtain power over private sector AI and retain power over future human-level AI. In summary, the combination of PoP consensus mechanisms, ZKP infrastructure and a blockchain-based AI shield form a cohesive

framework for democracy-driven, privacy-preserving and post-quantum resilient AI alignment.

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Declarations

Competing Interests The authors declare that they have no competing interests.

Ethical Approval Not applicable.

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